Masoud Aghayan

Data Scientist | Machine Learning & Business Analytics

Utrecht, Netherlands | +31 6 8212 4081 | masoud.aghayan64@gmail.com
LinkedIn | github.com/MasouData | Dutch citizen; no sponsorship required

Profile

Data scientist with an MSc in Computer Science and applied mathematics training, combining customer churn modelling, A/B test analysis and predictive analytics with Python, SQL and PySpark. Experienced in explaining results to business stakeholders and building tested analytical workflows. Currently works weekends at Albert Heijn in Zeist and seeks to contribute this technical background and store-level perspective to data science within the organisation.

Core Skills

Data science: Predictive modelling, classification, clustering, feature engineering, model evaluation, NLP.
Analysis: Python, SQL, PySpark, A/B testing, customer-retention analysis, Spotfire, stakeholder communication.
Reliable delivery: Azure Databricks, Delta Lake, ETL, data validation, Git, GitHub Actions, pytest.

Albert Heijn Experience

Current weekend store work *Albert Heijn, Emmaplein 2, Zeist*
Work weekends in the store alongside my technical career, gaining direct exposure to the retail environment.

Professional & Research Experience

Data Science & Analytics Engineer (Remote) Dec 2025–Present
AKO Pardaz, Tehran, Iran

- Developed Python churn models with feature engineering, model evaluation and threshold analysis; analysed A/B tests for retention, conversion and engagement to inform business recommendations.

Execution Anchor & Data Engineer Collaborator — TerraYield 2 Jul–Sep 2026
Omdena

- Collaborated on data pipelines and data-quality work; supported team coordination and delivery.

Data Scientist Intern — Predictive Analytics & NLP Feb–Sep 2025
ASML, Veldhoven, Netherlands

- Prepared operational data with Python, SQL and PySpark in Azure Databricks; applied schema validation, deduplication and documented transformations for reliable downstream analysis.
- Used Git, GitHub Actions and pytest to support reproducible, tested workflows; communicated analytical results and limitations to engineers and business controllers using Spotfire.
- Developed repair-versus-scrap prediction models with 83% held-out accuracy; built an NLP workflow for 12,000+ operational texts, reducing manual review time by approximately 60%.

Teaching Assistant — Python Programming & Process Analysis 2024–2025
Utrecht University, Netherlands

- Coached 50+ students in Python, debugging and validation, explaining analytical methods clearly.

Visiting Researcher — Law, Regulation & Machine Learning for AML 2020–2022
Tilburg University, Netherlands

- Studied interpretable and privacy-aware ML for anti-money laundering; interviewed Dutch compliance and legal professionals about practical requirements.

Data Analyst — Financial Risk, Fraud & AML Analytics 2015–2020
Mehr Iran Bank, Iran

- Analysed customer, transaction and repayment data for risk assessment and reporting; discussed findings with risk, audit and compliance stakeholders.
- Developed SQL ETL, reconciliation and recurring reporting workflows, reducing manual effort by 40%; investigated inconsistencies to improve the reliability of reporting data.

Selected Portfolio Projects

Semantic clustering | GitHub: Used sentence embeddings, UMAP and clustering to organise documents; added interpretable topic labels with KeyBERT and LLM assistance.

Streaming analytics pipeline | GitHub: Built a Kafka/Spark and Delta Lake Bronze/Silver/Gold pipeline with validation, deduplication, enrichment and analytical aggregates.

Education & Certifications

MSc Computer Science — Data Science, Utrecht University 2023–Sep 2025
BSc & MSc Applied Mathematics, Shahrood University of Technology Completed 2013
Microsoft Certified: Fabric Data Engineer Associate (DP-700) 28 Oct 2026
Databricks Certified Data Engineer Associate May 2026

Languages

English: fluent | Dutch: B1 (conversational)